Network Switch Types

Assignment 2: Comprehensive Technical Overview (Week 4)

Enterprise Switching Infrastructure

Architecture, Deployment Scenarios,
and Technology Specifications

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November 7, 2025

Network Switch Types - Comprehensive Overview

Executive Summary

Modern enterprise networks utilize various types of switches optimized for specific deployment scenarios and performance requirements. This document provides a comprehensive overview of different switch categories, their architectural roles, technical specifications, and appropriate use cases within multinational corporation environments.

Introduction

Network switches form the fundamental building blocks of enterprise network infrastructure, providing the connectivity fabric that interconnects devices across organizational boundaries. Different switch types are engineered to address specific performance, scalability, and feature requirements across campus, data center, and specialized environments. This document details the major switch categories deployed in modern MNC networks.

Part 1: Core Switch Categories

1.1 Core Switches

Function:
Core switches serve as the central backbone of enterprise networks, providing high-speed connectivity between different network segments and aggregation points. They handle the highest volume of traffic with maximum reliability and minimal latency, acting as the primary interconnection point for distribution layer switches, data centers, and internet gateways.

Key Characteristics:
Core switches feature extremely high port densities with support for 40G/100G/400G interfaces, non-blocking architectures that ensure full wire-speed performance across all ports, and advanced redundancy features including redundant power supplies, management modules, and fabric connectivity. They typically support extensive Layer 3 routing protocols including BGP, OSPF, and EIGRP for enterprise-scale routing.

Deployment Scenarios:

- **Enterprise Campus Backbone:** Interconnecting multiple distribution switches
- **Data Center Core:** Spine layer in spine-leaf architectures
- **Service Provider Edge:** Aggregating multiple customer networks

Technical Specifications:
Backplane capacity ranging from 10-100+Tbps, port densities of 48-144 ports of 100G or higher, support for advanced features like MPLS, VXLAN, and EVPN, with typical latency under 5 microseconds.

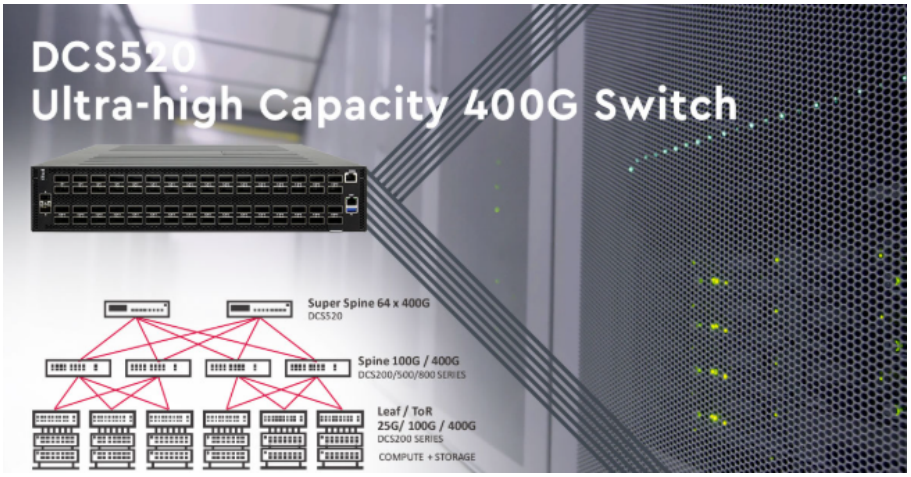

Figure 1: High-performance core switch with 100G/400G interface capabilities

1.2 Distribution Switches

Function:

Distribution switches operate as aggregation points between access layer switches and the network core, implementing policy enforcement, routing boundaries, and broadcast domain control. They provide the critical function of aggregating traffic from multiple access switches and applying centralized security and quality of service policies before forwarding to the core.

Key Characteristics:

Distribution switches feature high Layer 3 routing performance with support for inter-VLAN routing, policy-based routing, and access control lists. They implement advanced QoS mechanisms for traffic prioritization, provide redundancy through protocols like StackWise or VSS, and serve as the primary point for network segmentation and security policy enforcement.

Deployment Scenarios:

- **Campus Distribution:** Aggregating multiple access layer switches per building
- **Branch Office Core:** Serving as main switch for larger branch locations
- **Data Center Aggregation:** Connecting multiple access switches to core

Technical Specifications:

Typically 24-48 ports of 10G/25G with 4-8 uplink ports of 40G/100G, support for advanced features like sFlow/NetFlow for monitoring, IP SLA for performance measurement, and comprehensive security feature sets.

Figure 2: Distribution layer switch with high-density 10G/25G capabilities

1.3 Access Switches

Function:

Access switches provide the final network connectivity to end-user devices, including computers, phones, wireless access points, and IoT devices. They serve as the first point of entry to the network infrastructure, implementing port-level security, quality of service, and basic traffic segmentation.

Key Characteristics:

Access switches prioritize features like Power over Ethernet (PoE/PoE+) for powering connected devices, port security mechanisms including 802.1X authentication, and VLAN segmentation for separating user traffic. They typically operate at Layer 2 with optional Layer 3 capabilities, providing cost-effective connectivity while enforcing security policies at the network edge.

Deployment Scenarios:

- **Desktop Connectivity:** Connecting end-user computers and phones
- **Wireless Backhaul:** Connecting Wi-Fi access points
- **IoT Infrastructure:** Supporting connected devices and sensors

Technical Specifications:

24-48 ports of 1G/2.5G/5G with 2-4 uplink ports of 10G/25G, PoE+ budget of 370W-1100W, support for features like MACsec encryption, and energy-efficient Ethernet capabilities.

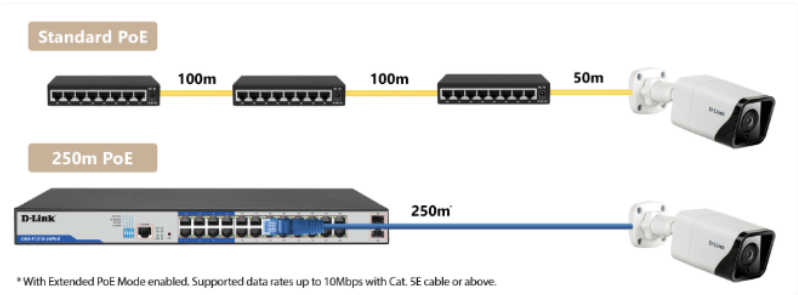

Figure 3: Access layer switch with PoE+ capabilities for end-device connectivity

1.4 Data Center Switches

Function:

Data center switches are specifically engineered for high-density server connectivity, storage networking, and application performance requirements. They form the fabric that interconnects servers, storage arrays, and application delivery controllers within data center environments.

Key Characteristics:

Data center switches feature extremely low latency (often sub-microsecond), high radix designs supporting large-scale leaf-spine architectures, and advanced features like VXLAN for network virtualization. They support lossless Ethernet for storage traffic (DCB), integrated analytics for visibility, and automation-friendly interfaces for orchestration.

Deployment Scenarios:

- **Leaf Switches:** Top-of-rack connectivity for server cabinets
- **Spine Switches:** Core fabric connectivity in leaf-spine architectures
- **Storage Networking:** Connecting SAN and NAS storage systems

Technical Specifications:

32-64 ports of 25G/100G with support for 400G uplinks, latency under 1 microsecond, large buffer sizes for burst absorption, and support for RDMA over Converged Ethernet (RoCE).

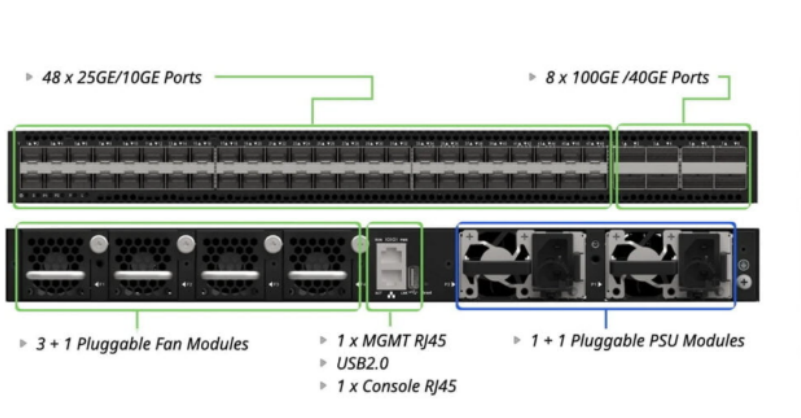

Figure 4: Data center leaf switch with high-density 25G/100G server connectivity

1.5 Managed vs. Unmanaged Switches

Function:

The fundamental distinction between managed and unmanaged switches determines the level of control, security, and monitoring capabilities available to network administrators. This choice represents the trade-off between operational simplicity and feature richness.

Key Characteristics:

Managed switches provide complete administrative control through web interfaces, CLI, or SNMP, supporting VLANs, QoS, port security, and monitoring features. Unmanaged switches operate as plug-and-play devices with zero configuration requirements, automatically negotiating connections but offering no management capabilities or security features.

Deployment Scenarios:

- **Managed Switches:** Enterprise networks, security-sensitive environments, networks requiring segmentation
- **Unmanaged Switches:** Small offices, home networks, temporary setups, non-critical applications

Technical Specifications:

Managed switches typically include feature sets for STP, LACP, ACLs, sFlow, and remote management, while unmanaged switches focus on basic connectivity with auto-negotiation and auto-MDI/X capabilities.

Figure 5: Enterprise managed switch with comprehensive management capabilities

1.6 Stackable Switches

Function:
Stackable switches provide a modular chassis-like functionality using individual fixed-configuration switches that are interconnected to operate as a single logical unit. This architecture offers the benefits of modular systems with the cost-effectiveness of fixed-configuration hardware.

Key Characteristics:
Stackable switches use dedicated stacking cables or ports to create a unified backplane across multiple physical devices, enabling single-point management, unified configuration, and cross-switch link aggregation. They provide hitless failover within the stack, simplified software upgrades across the stack, and flexible scalability by adding units as needed.

Deployment Scenarios:
- **Branch Offices:** Providing modular capabilities in space-constrained environments
- **Campus Access:** Scaling port density beyond single switch limitations
- **Mid-Size Data Centers:** Creating cost-effective fabric architectures

Technical Specifications:
Stacking bandwidth of 40G-480G per member, support for 4-12 units per stack, unified management interface, and hitless software upgrades across stack members.

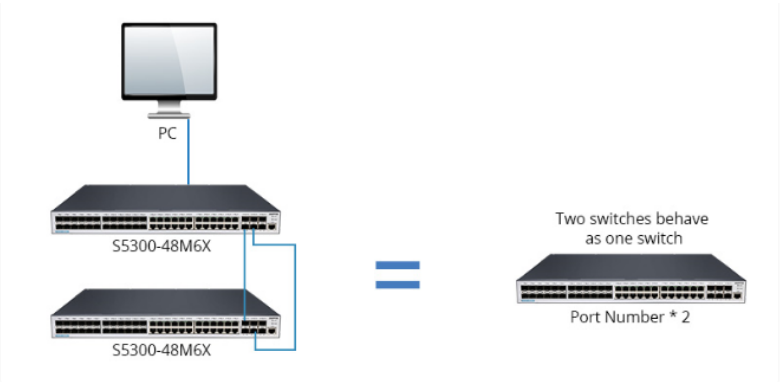

Figure 6: Stackable switch system operating as single logical unit

1.7 Modular Chassis Switches

Function:

Modular chassis switches provide the highest levels of scalability, redundancy, and service integration in enterprise networks. They consist of a chassis with multiple slots that accept various line cards, supervisor engines, and power supplies, enabling customized configurations for specific requirements.

Key Characteristics:

Modular switches offer complete service isolation, with separate control plane (supervisor), data plane (line cards), and power/cooling systems. They provide N+1 or N+N redundancy for all components, hot-swappable field replacement units, and the ability to mix different interface types and technologies within a single chassis.

Deployment Scenarios:

- **Network Core:** High-density aggregation and core routing
- **Data Center Aggregation:** Multiple technology integration points
- **Service Provider:** Multi-service edge and peering points </endkeypoints

Specifications: Chassis sizes from 4-20 slots, backplane capacity from 1-100+Tbps, support for mixed 1G/10G/25G/40G/100G/400G line cards, and complete component redundancy.

Figure 7: Modular chassis switch with redundant supervisors and line cards

Strategic Selection Guidelines

Switch Selection Framework

- **Performance Requirements:** Evaluate throughput, latency, and buffer size based on application needs

- **Scalability Planning:** Consider future growth requirements and expansion capabilities
- **Feature Requirements:** Identify necessary Layer 2/3 features, security, and management capabilities
- **Total Cost of Ownership:** Analyze acquisition costs, power consumption, cooling, and operational expenses
- **Operational Simplicity:** Balance feature richness with management complexity based on staff expertise